

CONTACT
INFORMATION

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EDUCATION

- University of Houston**, College of Natural Sciences and Mathematics Aug 2022–Present
 - Research Assistant, PhD Candidate
 - M.Sc. Physics, GPA: **3.702** Aug 2022–Dec 2025
Advisor: Prof. Eric R. Bittner
- Durham University**, Ustinov College, Durham, UK Oct 2018– Sept 2019
 - M.Sc. in [Particles, Strings and Cosmology](#), Result: **Merit** – 180 credits.
Thesis: **An Introduction to AdS/CFT and the Holographic Entanglement Entropy.**
Durham University, UK
Supervisor: Prof. Simon F. Ross, Reader: Prof. Nabil Iqbal
- Amity University Noida**, India. Aug 2015–May 2018
 - B.Sc. (Honours) [Physics](#), CGPA: **8.05/10 (First Division)** – 176 credits.

MOST RECENT
PUBLICATIONS

3. G. Valdivia-Mera, **B. Tyagi**, E. R. Bittner and P. Hosur. “The Helical SYK Model and Emergent Infrared Integrability” [arXiv](#) 2026
2. E. R. Bittner and **B. Tyagi**. “Clockwork quantum: symmetry, noise, and the emergence of quantum order” [Philosophical Magazine](#) 2025
1. **B. Tyagi**, F. Suzuki, V. A. Chernyak, and N. A. Sinitsyn. “Asymmetry Amplification by a Nonadiabatic Passage through a Critical Point” [Physical Review A](#) 2025

The full list can be found on my [Google Scholar](#) or [arXiv](#).

CURRENT
RESEARCH

- **PhD Thesis: Noise-induced Order in Quantum Matter** January 2024–Present
University of Houston, Texas, USA
PhD Advisor: Prof. Eric R. Bittner
- **Entanglement Dynamics in the Kicked-Field Ising Model** December 2025–Present
Collaborators: Prof. Tomaž Prosen, Dr. Tanay Pathak
- **Helical SYK Model in 1+1 Dimension** June 2025–June 2026
Collaborator (Thesis Committee Member): Prof. Pavan Hosur
- **Majorana Hub Models** June 2024–Present
Collaborator (Thesis Committee Member): Dr. Nikolai A. Sinitsyn

ACHIEVEMENTS

- European Cooperation in Science and Technology Fellowship 2025
- APS Science Advocacy Champion Award 2025
- Cullen Fellowship, University of Houston 2025
- Best Talk, Physics Research Day 2025, University of Houston 2025
- Graduate Research Fellow at *Los Alamos National Lab* 2024
- Awarded Distinction for Masters Thesis 2020
- Undergraduate Research Fellow *Harish Chandra Research Institute* 2017

LEADERSHIP & TEACHING EXPERIENCE	• American Physical Society Advocacy Champion	2025–Present
	• Chair American Physical Society (APS) Chapter at University of Houston	2024–2025
	• Physics 2125 (University Level Classical Mechanics)	2023–2024
	• Physics 1101 (College Level Classical Mechanics)	2022–2023
	• Physics 2126 (Topics in Modern Physics: Waves, Quantum Mechanics, Nuclear Physics)	Summer 2023
TALKS/POSTERS		
†Invited		
	12. Talk Title: Are we living in a Matrix? [†] (Public Talk) After 5 Society, Patterson Park <i>Houston, Texas</i>	April 2026
	11. Talk Title: Lessons from Fundamental Physics [†] (Public Talk) Sanskriti School <i>New Delhi, India</i>	January 2026
	10. Talk Title: On “Noise-Friendly” Quantum Systems [†] Indian Institute of Technology Delhi, India	January 2026
	9. Talk Title: On “Noise-Friendly” Quantum Systems Telluride Science and Research Center <i>Telluride, Colorado, USA</i>	September 2025
	8. Poster Title: Asymmetry Amplification by a Non-Adiabatic Passage Through a Critical Point University of Warsaw <i>Warsaw, Poland</i>	September 2025
	7. Poster Title: Noise-Induced Synchronisation and Entanglement APS Global Physics Summit <i>Anaheim, California, USA</i>	March 2025
	6. Talk Title: Asymmetry Amplification by a Non-Adiabatic Passage through a Critical Point APS Global Physics Summit <i>Anaheim, California, USA</i>	March 2025
	5. Talk Title: On “Noise-Friendly” Quantum Systems University of Houston Physics Research Day 2025 <i>Houston, Texas, USA</i>	February 2025
	4. Talk Title: Asymmetry Amplification by a Non-Adiabatic Passage through a Critical Point CNLS Summer Student Talks <i>Center for Non-Linear Studies, Los Alamos National Lab, NM, USA</i>	Summer 2024
	3. Talk Title: Black Hole Information Problem & Recent Developments [†] Quantum Photonics Physics Forum [†] (Public Talk) <i>Online</i>	September 2021
	2. Talk Title: On Quantum Entanglement and the Interpretation of Quantum Mechanics. [†] (Public Talk) Christ University Physics Club <i>Christ University Bangalore, India</i>	2020
	1. Talk Title: On Relativity and Gravitation: Applications to Modern Physics. (Public Talk) Beyond Portals Lecture Series <i>Beyond Portals HQ, New Delhi, India</i>	2018
EXTRA- CURRICULAR ACTIVITIES	• The Knowmads Podcast	
	• Music Channel	
	• Science Advocacy and Policy Making	